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15:53:23 ~/$SJSU_MSADI/FALL_2026/F2026_DATA_226/HOMEWORK/HOMEWORK_4/hw4 dbt debug
15:53:23 Running with dbt=1.12.5
15:53:23 dbt version: 1.12.5
15:53:23 python version: 3.13.13
15:53:23 python path: /Users/jaypas/py/bin/python3.13
15:53:23 os info: macOS-15.7.4-arm64-arm-64bit-Mach-O
15:53:23 Using profiles dir at /Users/jaypas/.dbt
15:53:23 Using profiles.yml file at /Users/jaypas/.dbt/profiles.yml
15:53:23 Using dbt_project.yml file at /Users/jaypas/SJSU_MSADI/FALL_2026/F2026_DATA_226/HOMEWORK/HOMEWORK_4/hw4/dbt_project.yml
15:53:23 adapter type: snowflake
15:53:23 adapter version: 1.12.1
15:53:23 Configuration:
15:53:23   profiles.yml file [OK found and valid]
15:53:23   dbt_project.yml file [OK found and valid]
15:53:23 Required dependencies:
15:53:23   - git [OK found]

15:53:23 Connection:
15:53:23   account: QBWCWAW-KY14417
15:53:23   user: JAYPASNAGASAI
15:53:23   database: DEMO_DB
15:53:23   warehouse: COMPUTE_WH
15:53:23   role: ACCOUNTADMIN
15:53:23   schema: PUBLIC
15:53:23   authenticator: None
15:53:23   oauth_client_id: None
15:53:23   query_tag: None
15:53:23   client_session_keep_alive: False
15:53:23   host: None
15:53:23   port: None
15:53:23   proxy_host: None
15:53:23   proxy_port: None
15:53:23   protocol: None
15:53:23   connect_retries: 1
15:53:24   connect_timeout: None
15:53:24   retry_on_database_errors: False
15:53:24   retry_all: False
15:53:24   insecure_mode: False
15:53:24   reuse_connections: True
15:53:24   s3_stage_vpce_dns_name: None
15:53:24   workload_identity_provider: None
15:53:24   workload_identity_entra_resource: None
15:53:24   platform_detection_timeout_seconds: 0.0
15:53:24 Registered adapter: snowflake=1.12.1
/Users/jaypas/py/lib/python3.13/site-packages/snowflake/connector/config_manager.py:409: UserWarning: Bad owner or permissions on /Users/jaypas/.snowflake/connections.toml.
* To change owner, run `chown $USER "/Users/jaypas/.snowflake/connections.toml"`.
* To restrict permissions, run `chmod 0600 "/Users/jaypas/.snowflake/connections.toml"`.
* To skip this warning, set environment variable SF_SKIP_TOKEN_FILE_PERMISSIONS_VERIFICATION=true.

    warn(f"Bad owner or permissions on {str(filep)}{chmod_message}")
15:53:27 Connection test: [OK connection ok]

15:53:27 All checks passed!
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```

1  -- USER SESSION SQL FILE-
2
3
4  {{ config(materialized='table') }}
5
6  WITH sessions AS (
7      SELECT-
8          session_id,
9          user_id,
10         start_time,
11         end_time,
12         device_type,
13         referrer
14     FROM {{ source('raw', 'sessions') }}
15     WHERE start_time IS NOT NULL
16 ),
17
18 users AS (
19     SELECT-
20         user_id,
21         username,
22         email,
23         signup_date,
24         country
25     FROM {{ source('raw', 'users') }}
26 ),
27
28 session_with_channel AS (
29     SELECT-
30         s.session_id,
31         s.user_id,
32         u.username,
33         u.email,
34         u.country,
35         s.device_type,
36         CASE-
37             WHEN LOWER(s.referrer) LIKE '%google%' THEN 'organic_search'
38             WHEN LOWER(s.referrer) LIKE '%facebook%' THEN 'social'
39             WHEN LOWER(s.referrer) LIKE '%direct%' THEN 'direct'
40             ELSE 'other'
41         END AS channel,
42         s.start_time,
43         s.end_time
44     FROM sessions s
45     LEFT JOIN users u ON s.user_id = u.user_id
46 ),
47
48 final AS (
49     SELECT-
50         session_id,
51         user_id,
52         username,
53         email,
54         country,
55         device_type,
56         channel,
57         start_time,
58         end_time,
59         CURRENT_TIMESTAMP() AS dbt_created_at
60     FROM session_with_channel
61     WHERE session_id IS NOT NULL
62 )
63
64 SELECT * FROM final
65
66

```

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1  -- SESSION TIMESTAMPS SQL FILE-
2
3  {{ config(materialized='table') }}
4
5  WITH sessions AS (
6      SELECT
7          session_id,
8          user_id,
9          start_time,
10         end_time
11     FROM {{ source('raw', 'sessions') }}
12     WHERE start_time IS NOT NULL
13 ),
14
15 timestamps_transformed AS (
16     SELECT
17         session_id,
18         user_id,
19         start_time,
20         end_time,
21         DATE(start_time) AS session_date,
22         YEAR(start_time) AS session_year,
23         MONTH(start_time) AS session_month,
24         DAYOFWEEK(start_time) AS session_day_of_week,
25         HOUR(start_time) AS session_hour,
26         DATEDIFF(MINUTE, start_time, end_time) AS session_duration_minutes,
27         CONCAT(YEAR(start_time), '-W', LPAD(WEEK(start_time), 2, '0')) AS session_week_id
28     FROM sessions
29 ),
30
31 sessions_with_labels AS (
32     SELECT
33         *,
34         CASE
35             WHEN session_hour BETWEEN 6 AND 11 THEN 'morning'
36             WHEN session_hour BETWEEN 12 AND 17 THEN 'afternoon'
37             WHEN session_hour BETWEEN 18 AND 23 THEN 'evening'
38             ELSE 'night'
39         END AS time_of_day,
40         CASE
41             WHEN session_day_of_week IN (1, 7) THEN 'weekend'
42             ELSE 'weekday'
43         END AS day_type
44     FROM timestamps_transformed
45 ),
46
47 final AS (
48     SELECT
49         session_id,
50         user_id,
51         start_time,
52         end_time,
53         session_date,
54         session_year,
55         session_month,
56         session_day_of_week,
57         session_hour,
58         session_week_id,
59         session_duration_minutes,
60         time_of_day,
61         day_type
62     FROM sessions_with_labels
63 )
64
65 SELECT * FROM final

```

```

3  {{ config(
4      materialized='table',
5      tags=["analytics", "core"]
6  ) }}
7
8  WITH user_sessions AS (
9      SELECT * FROM {{ ref('user_session_channel') }}
10 ),
11
12 session_times AS (
13     SELECT * FROM {{ ref('session_timestamp') }}
14 ),
15
16 combined_sessions AS (
17     SELECT
18         us.session_id,
19         us.user_id,
20         us.username,
21         us.email,
22         us.country,
23         us.device_type,
24         us.channel,
25         us.start_time,
26         us.end_time,
27         st.session_date,
28         st.session_year,
29         st.session_month,
30         st.session_day_of_week,
31         st.session_hour,
32         st.session_week_id,
33         st.session_duration_minutes,
34         st.time_of_day,
35         st.day_type
36     FROM user_sessions us
37     INNER JOIN session_times st
38         ON us.session_id = st.session_id
39 ),
40
41 session_summary_agg AS (
42     SELECT
43         session_id,
44         user_id,
45         username,
46         email,
47         country,
48         device_type,
49         channel,
50         session_date,
51         session_year,
52         session_month,
53         session_week_id,
54         session_day_of_week,
55         session_hour,
56         time_of_day,
57         day_type,
58         start_time,
59         end_time,
60         session_duration_minutes,
61         1 AS session_count,
62         CASE WHEN session_duration_minutes > 0 THEN 1 ELSE 0 END AS active_session_flag
63     FROM combined_sessions
64     WHERE session_id IS NOT NULL
65 ),
66
67 final AS (
68     SELECT
69         session_id,
70         user_id,
71         username,
72         email,
73         country,
74         device_type,
75         channel,
76         session_date,
77         session_year,
78         session_month,
79         session_week_id,
80         session_day_of_week,
81         session_hour,
82         time_of_day,
83         day_type,
84         start_time,
85         end_time,
86         session_duration_minutes,
87         session_count,
88         active_session_flag,
89         CURRENT_TIMESTAMP() AS dbt_created_at,
90         CURRENT_TIMESTAMP() AS dbt_updated_at
91     FROM session_summary_agg
92 )
93
94 SELECT * FROM final

```

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1  -- SNAPSHOT SESSION SUMMARY SQL FILE-
2
3  {% snapshot snapshot_session_summary %}
4
5      {% set build_incremental_logic = execute %}
6
7      {{ config(
8          target_schema='snapshots',
9          unique_key='session_id',
10         strategy='timestamp',
11         updated_at='dbt_updated_at',
12         tags=['snapshots', 'session_summary']
13     ) }}
14
15     SELECT-
16         session_id,
17         user_id,
18         username,
19         email,
20         country,
21         device_type,
22         channel,
23         session_date,
24         session_year,
25         session_month,
26         session_week_id,
27         session_day_of_week,
28         session_hour,
29         time_of_day,
30         day_type,
31         start_time,
32         end_time,
33         session_duration_minutes,
34         session_count,
35         active_session_flag,
36         dbt_created_at,
37         dbt_updated_at
38     FROM {{ ref('session_summary') }}
39
40 {% endsnapshot %}

```



